

PILATUS, Switzerland

WMO: 066590

Lat: 46.9789N

Lon: 8.2522E

Elev: 2110

StdP: 78.42

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.3	-14.3	-30.9	0.3	-2.1	-27.1	0.4	-3.6	20.2	-1.8	17.7	-3.9	6.3	40	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	18.7	12.4	17.2	11.8	15.9	11.1	13.9	16.9	13.0	15.8	12.1	14.8	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.8	15.3	11.7	11.1	14.4	10.8	10.5	13.6	46.3	17.1	43.5	15.9	40.8	14.9	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.9	11.9	DB	-18.1	20.8	2.9	1.7	-20.1	22.0	-21.8	23.0	-23.4	24.0	-25.5	25.2	(4)
(5)			WB	-18.7	15.8	2.6	1.0	-20.6	16.5	-22.1	17.0	-23.6	17.6	-25.5	18.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-4.2	-4.7	-2.3	0.9	3.9	8.3	9.8	9.9	6.6	4.7	0.1	-2.5	
	DBStd	6.89	4.44	5.37	4.87	4.45	4.42	4.49	3.98	4.20	4.17	4.75	5.00	4.71	
	HDD10.0	2864	442	411	380	272	196	84	53	56	114	171	298	388	
	HDD18.3	5748	700	644	638	522	448	301	264	262	351	423	548	646	
	CDD10.0	158	0	0	0	0	6	33	48	52	14	6	0	0	
	CDD18.3	1	0	0	0	0	0	1	1	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	5.4	6.1	6.1	5.7	5.4	5.2	4.7	4.4	4.1	4.9	5.7	6.3	6.4	
Precipitation	PrecAvg	1646	94	105	109	126	175	183	183	188	136	107	125	116	
	PrecMax	2069	224	307	216	253	278	283	348	407	233	190	346	289	
	PrecMin	1252	5	33	43	28	42	109	78	56	54	4	6	5	
	PrecStd	186	56	62	49	51	70	44	67	73	45	45	68	63	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	8.2	9.1	11.5	15.8	20.8	20.4	20.3	16.3	15.3	11.2	8.2	
		MCWB	0.1	0.3	1.8	5.4	9.8	13.1	13.7	12.6	9.7	7.4	3.2	0.2	
	2%	DB	5.0	6.4	6.9	9.8	13.4	18.0	18.2	18.6	14.6	13.2	9.4	6.5	
		MCWB	-1.0	-0.9	0.9	4.8	8.4	11.9	12.5	12.2	9.2	6.2	2.8	-0.9	
	5%	DB	3.2	3.9	5.4	8.4	11.9	16.2	16.7	17.1	13.3	11.9	7.9	5.0	
		MCWB	-1.7	-1.4	-0.1	3.8	7.5	11.5	11.9	11.7	8.8	5.6	2.1	-1.0	
	10%	DB	1.6	1.9	3.9	6.9	10.2	14.5	15.3	15.7	12.2	10.5	6.6	3.7	
		MCWB	-2.1	-2.5	-0.5	2.8	6.6	10.5	11.2	11.1	8.1	4.9	1.3	-1.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.0	3.5	7.1	11.3	15.1	15.3	15.0	12.2	9.1	5.5	2.5
			MCDB	4.1	5.6	6.4	9.9	14.5	18.7	19.0	17.7	14.4	12.1	8.8	4.4
2%		WB	0.9	0.4	2.2	5.7	9.6	13.5	13.7	13.7	10.9	7.8	4.1	1.2	
		MCDB	2.4	3.1	4.8	8.7	12.2	16.4	16.7	16.7	13.1	10.8	7.0	3.3	
5%		WB	-0.2	-0.6	1.2	4.5	8.1	12.3	12.7	12.8	9.9	7.0	3.2	0.2	
		MCDB	1.5	2.3	3.8	7.4	11.0	15.0	15.5	15.7	12.1	10.2	6.1	2.9	
10%		WB	-1.4	-1.6	0.2	3.4	6.9	11.1	11.8	11.8	8.9	6.1	2.2	-0.5	
		MCDB	0.9	1.5	2.8	6.3	9.4	13.6	14.4	14.6	11.3	9.2	5.5	2.3	
Mean Daily Temperature Range		5% DB	MDBR	5.1	5.1	5.2	5.0	5.0	5.0	5.1	4.9	4.8	5.2	5.0	5.1
			MCDBR	5.9	6.2	6.1	5.9	6.3	6.0	6.2	6.0	5.6	6.0	5.9	5.9
	MCWBR		3.9	4.0	4.3	4.9	5.6	5.8	5.5	5.8	5.5	4.4	3.7	3.6	
	5% WB	MCDBR	5.6	5.8	5.4	5.8	6.0	5.6	5.8	5.5	5.0	5.4	5.4	5.7	
		MCWBR	5.1	4.5	4.2	5.0	5.5	5.8	5.6	5.7	5.3	4.4	4.2	4.7	
Clear-Sky Solar Irradiance	taub	0.227	0.247	0.278	0.307	0.320	0.324	0.325	0.318	0.298	0.279	0.250	0.225		
	taud	2.452	2.420	2.423	2.415	2.424	2.442	2.423	2.460	2.512	2.555	2.550	2.469		
	Ebn at Noon	904	946	955	951	944	938	932	928	922	896	867	867		
	Edh at Noon	63	82	96	107	111	110	110	102	88	72	58	55		
All-Sky Solar Radiation	RadAvg	1.35	2.24	3.44	4.39	4.91	5.74	5.62	4.89	3.75	2.42	1.40	1.11		
	RadStd	0.16	0.29	0.49	0.67	0.48	0.53	0.50	0.47	0.37	0.27	0.19	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (119 neighbors)	N/A	N/A	N/A	+0.72	+0.40	+0.31	N/A	N/A	+57	+15

Nomenclature: See separate page